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Two-Step Improvement of Thermoelectric Properties of PEDOT:PSS Films by Femtosecond Laser and Ammonia Vapor Atmosphere Treatment

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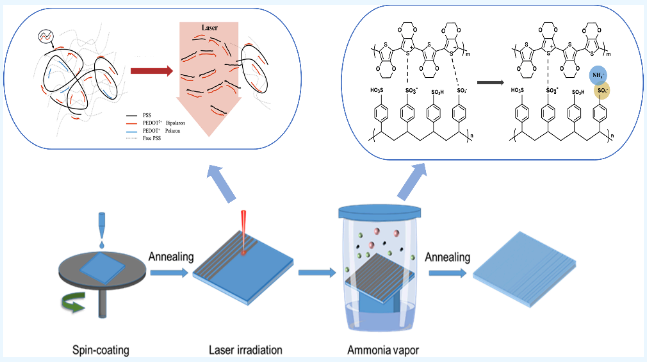

ABSTRACT: Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonic acid) (PEDOT:PSS) exhibits exceptional optical properties and flexibility. Additionally, its aqueous suspension demonstrates commercial viability, making this material particularly suitable for low-cost solution processing applications for flexible electronic devices. However, the relatively low initial conductivity (σ) of PEDOT:PSS (~1 S cm^{-1}) makes it unsuitable for direct application in flexible thermoelectric devices. Herein, we report a two-step approach to improve the thermoelectric properties of PEDOT:PSS films by a femtosecond laser and ammonia vapor atmosphere treatment. First, femtosecond laser treatment reduces the PSS content and causes the benzoquinone conformational transformation of the PEDOT molecular chain. Although it increases the electrical conductivity (σ), it leads to a decrease in the Seebeck coefficient (S). The subsequent ammonia vapor atmosphere treatment precisely regulates the PEDOT dedoping level, utilizes the binding effect between NH_4^+ and PSS^- to weaken the interaction force between the two, enhances the S, and thereby enhances the overall thermoelectric performance. The treated PEDOT:PSS films showed S, σ, and corresponding power factor (PF) values of 26.15 μV K^{-1}, 413.19 S cm^{-1}, and 28.24 $\mu W \cdot m^{-1} \cdot K^{-2}$, respectively. The prepared organic thermoelectric device shows an output voltage of 5.3 mV at a temperature difference of 40.6 °C.

KEYWORDS: *PEDOT:PSS, Femtosecond laser irradiation, Ammonia atmosphere, Thermoelectric device, Conducting polymers*

1. INTRODUCTION

Thermoelectric (TE) materials, as an emerging energy material, are a promising class of candidate materials capable of converting low-grade waste heat that cannot be fully exploited into electricity on a large scale.[1−4] Inorganic TE materials such as Bi_2Te_3, Sb_2Te_3, GeTe, and PbTe have been widely used due to their good thermoelectric characteristics.[5−8] However, their applications are limited due to their high cost, toxicity, and potential for environmental damage. Compared to inorganic materials, organic thermoelectric materials such as PEDOT:PSS have low toxicity, low cost, and inherently low thermal conductivity, which has sparked significant interest due to their excellent thermal stability, high flexibility, and ability to modulate their properties via doping.[9−12] The performance of thermoelectric materials is typically evaluated using the dimensionless figure of merit (ZT), defined as $ZT = S^2 \cdot \sigma \cdot T/\kappa$, where σ, S, T, and κ represent electrical conductivity, Seebeck coefficient, absolute temperature, and thermal conductivity (κ), respectively.[13−15] We can also evaluate the thermoelectric performance of PEDOT:PSS thin films using the power factor (PF), where $PF = S^2\sigma$. Here, σ and S represent the electrical conductivity and Seebeck coefficient.[16] For the PEDOT:PSS conductive polymer, improved conductivity and Seebeck coefficient lead to better TE energy efficiency.

The main methods for optimizing the PF of conductive polymers are chemical and physical doping.[17−23] However, the use of dopants in treatment can easily lead to degradation of film quality, diminishing the competitiveness of flexible films in the field of thermoelectrics.[24−26] Moreover, it inevitably involves the use or generation of harmful chemicals, which pose risks to human health and the environment. While physical doping methods can modify the σ of PEDOT:PSS,

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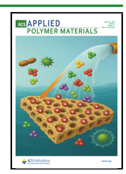

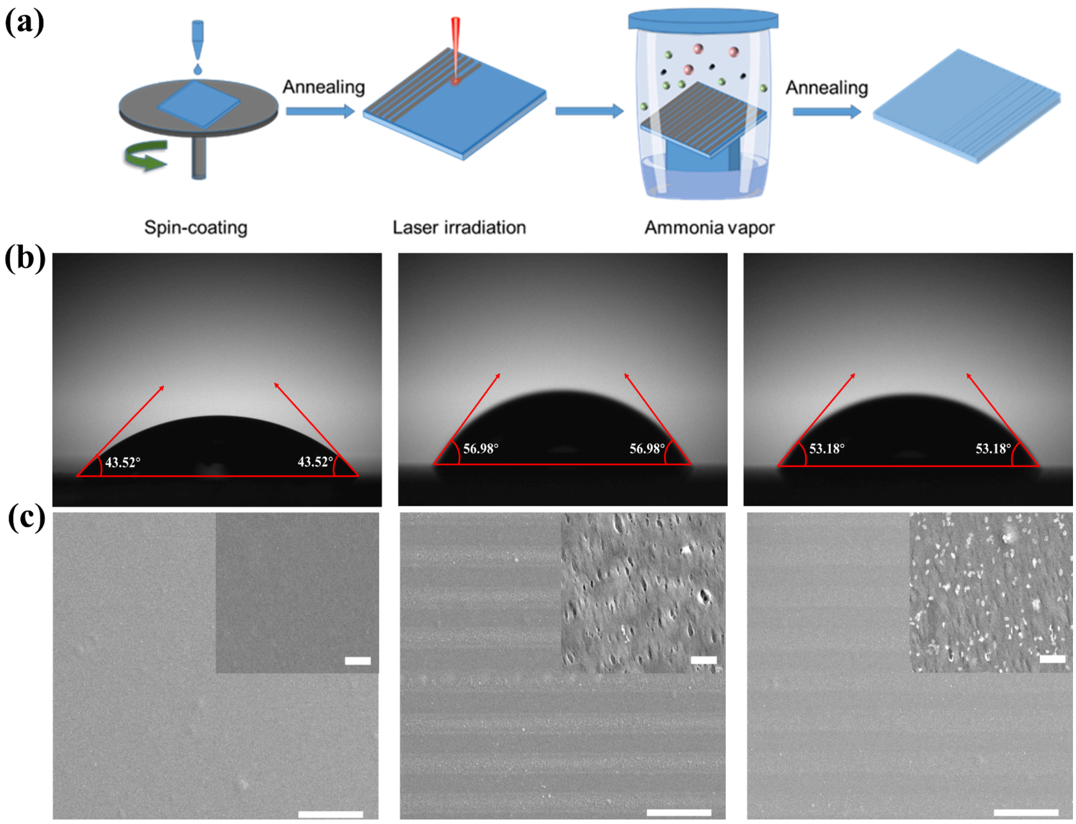

Figure 1. (a) Flowchart of two-step preparation of PEDOT:PSS thermoelectric films by femtosecond laser irradiation and AV atmosphere treatment. (b) Water droplet contact angles on pristine, laser-treated, and laser and AV atmosphere treated PEDOT:PSS films. (c) Morphology of PEDOT:PSS films with the pristine sample, the laser treatment, and the AV atmosphere treatment after laser irradiation. Scale bar: 50 μm. The inset shot scale: 1 μm.

they usually cannot achieve synergistic optimization with S. Additionally, certain physical doping approaches may compromise the inherent flexibility of PEDOT:PSS, limiting its applicability to flexible electronic devices. In previous work, we innovatively proposed a femtosecond (fs) laser annealing process to develop high-performance thermoelectric thin films with high performance.[27,28] Femtosecond laser processing of organic thin films is characterized by noncontact, no pollution, high processing efficiency, and high precision.[29] It can also regulate the electronic structure of polymer thin films.[30] The insulating PSS component was partially eliminated by the irradiation effect of the femtosecond laser. In addition, the stacking distance of the PEDOT chain was reduced by laser stamping. Therefore, enhancing the transport capability increases the carrier concentration, leading to an increase in the σ. During this process, the original PEDOT:PSS film PF can be increased to 18.81 $\mu W \cdot m^{-1} \cdot K^{-2}$, an improvement of 2 orders of magnitude. However, the S of the film was adversely affected, and the thermoelectric properties of the film could not be improved to the optimum. Alkali vapor treatment utilizes a noncontact processing method, effectively avoiding damage to the PEDOT:PSS film. Alkali vapor treatment enables a rapid and uniform addition process. This method resulted in a denser and more uniform film after treatment. Additionally, alkali vapor treatment can adjust the doping level of molecular chains, thereby enhancing S; however, it does not simultaneously achieve a significant improvement in conductivity.[31−33]

In this study, we propose a two-step method combining femtosecond laser annealing and ammonia atmosphere treatment to improve charge transfer and accurately control doping levels simultaneously to enhance the thermoelectric properties of PEDOT: PSS films. Irradiation with a femtosecond laser partially removed the insulating PSS and reduced the stacking distance of the PEDOT chains. The synergistic effect of the optimal dedoping regulated by treatment in an ammonia vapor atmosphere leads to an optimized carrier concentration. The two-step method is green and environmentally friendly. The two-step method collaboratively achieved the synchronous optimization of σ and S. The maximum S and σ of the PEDOT:PSS films were 26.15 $\mu V\ K^{-1}$ and 413.19 $S\ cm^{-1}$, respectively, while the corresponding PF was 28.24 $\mu W \cdot m^{-1} \cdot K^{-2}$. This combined approach resulted in an optimized carrier concentration. To identify the advantages of the proposed method, organic thermoelectric devices were fabricated using PEDOT:PSS films treated with the two-step method, and the output voltage of the devices was 5.3 mV when the temperature difference was 40.6 °C. This demonstrates the potential of the two-step method combining femtosecond laser treatment and ammonia atmosphere treatment in the preparation of flexible thermoelectric integrated devices.

2. MATERIAL AND METHODS

2.1. Materials. PEDOT:PSS (Clevios PH1000, 1.3 wt %) was purchased from Heraeus. NH_3-H_2O was purchased from Sinopharm.

2.2. Femtosecond Laser Treatment of PEDOT:PSS Films. As shown in Figure 1, PEDOT:PSS films with a thickness of about 100 nm were prepared by spin-coating on glass and polyethylene terephthalate (PET) substrates. Subsequently, the films were irradiated by the laser galvo scanner processing system with the

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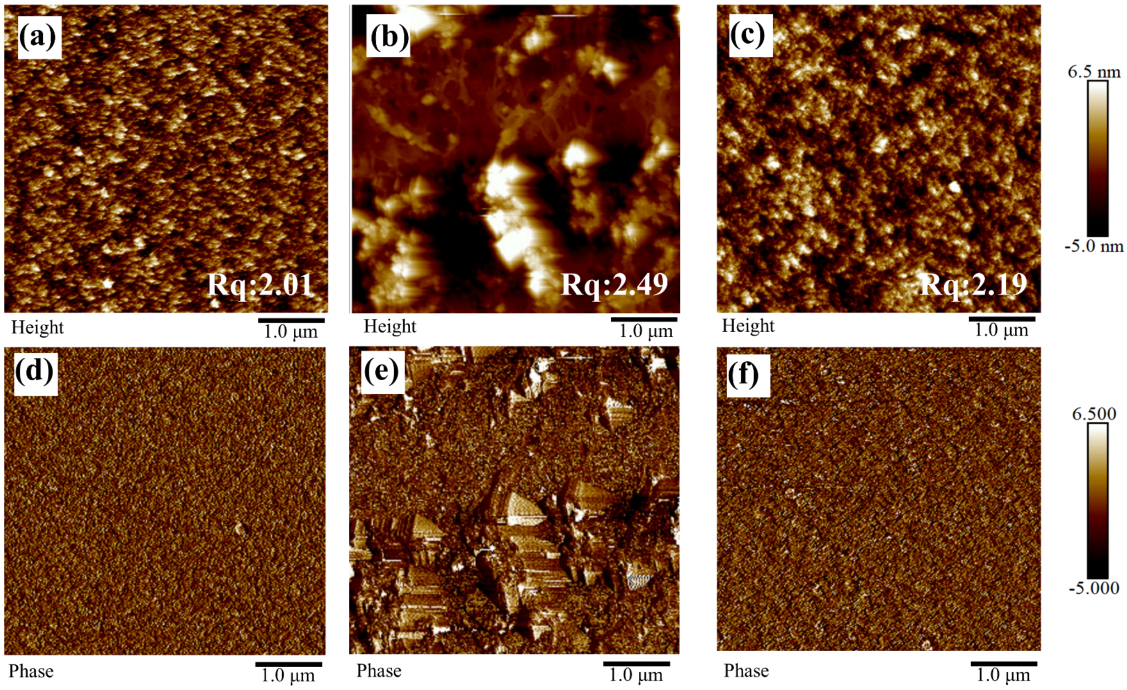

Figure 2. AFM images of morphology maps for (a) pristine, (b) laser treated, and (c) laser and AV atmosphere treated PEDOT:PSS films. The corresponding phase maps of (d) pristine, (e) laser-treated, and (f) laser and AV atmosphere-treated PEDOT:PSS films. All the scale bars are 1 μm.

following parameters: laser wavelength of 1030 nm, scanning distance of 7 μm, pulse width of 260 fs, repetition frequency of 200 kHz, spot radius of 12.5 μm, number of pulses of 15, processing speed of 300 mm s^{-1}, and laser fluence density of 58.08 mJ cm^{-2} (corresponding to a laser power of 58 mW).

2.3. Ammonia Atmosphere Treatment. 50 mL ammonia solution with different volume molar concentration of mol L^{-1} (M) is configured, in which the concentration of ammonia−water is 0.01 0.1, 0.2, 0.3, and 0.4 M. The polymer film treated with femtosecond laser was removed. The laser-treated PEDOT:PSS film was then secured to the experimental device rack with the substrate, and the experimental device rack containing the film samples was placed in an upright position in a closed specific glass vial (Figure 1). 5 mL portion of ammonia solution of different volumetric molar concentrations was added to the specific glass vials, which were later heated in an oven at 60 °C for 0−60 min. At the end of the treatment time, 100 μL of deionized water was dropped on the films for cleaning. Afterward, the films were placed on a spin coater at 2000 rmp for 20 s to remove the aqueous solution on the surface of the films. The annealing process was carried out on a hot plate at 80 °C for 15 min.

2.4. Measurement and Characterization. The thermal electrical properties of the film were measured with the CTA-3/500 model thermal electrical test system of Cree Europe. The thickness of the sample was obtained by using a DektakXT profilometer. The surface morphology of the film was obtained by using a NanoScopeIV atomic force microscope (AFM). The X-ray diffraction (XRD) measurement of the film was carried out by using a Bruker D8-Advance X-ray diffractometer to analyze the crystallinity and crystal plane orientation of the sample. The molecular structure and chemical composition of the film were determined by using a Nicolet 6700 device. The microstructure of the PEODT:PSS samples were obtained using a JSM-IT800 SEM. Raman spectroscopy (Raman) was used as a nondestructive testing method to provide information on the chemical structure, relative content, and crystallinity of the sample.

2.5. Device Fabrication and Measurement. This thermoelectric (TE) device is composed of thermoelectric legs made of PEDOT:PSS films treated by a two-step method, and the ends of the thermoelectric legs are connected by conductive silver paste. Polyethylene (PE) was adopted as the substrate, and six 30 mm × 5 mm PEDOT:PSS/AV films were used as the power generation units. It is composed of six thermoelectric legs and adhered to the polyethylene (PE) film. Place one side of the prepared TE device at the end with the heating platform as the heat source and the other side as the cold source. After the temperatures on both sides of the thermoelectric legs stabilize, measure the temperature difference on both sides through infrared and obtain the output voltage through formula 4. Then place one side of the TE device on the hot table and the other side in the environment. By adjusting the temperature of the hot stage to control the temperature difference at both ends of the TE device, the output voltage of the TE device under different temperature differences can be tested

3. RESULTS AND DISCUSSION

Characterization techniques such as UV−vis−NIR, Raman, and XRD were used. We found that femtosecond laser irradiation of polymer films could cause oxidation of PEDOT, simultaneously leading to a decrease in the content of insulating PSS in the films and promoting the transformation of thiophene rings in PEDOT from benzene-like to quinone-like structures. This promoted the structural reorganization of PEDOT:PSS from the core−shell model to a rod-shaped model. Moreover, the $\pi-\pi$ stacking distance of PEDOT was reduced, and the crystallinity of the molecular chain was increased, thereby enhancing the carrier transport capacity (Figures S1−S4). However, the S of the film was adversely affected, and the thermoelectric properties of the film could not be improved to the optimum (Figure S5). As shown in Figure 1a, to improve the thermoelectric properties, we treated the films with an ammonia vapor (AV) atmosphere after femtosecond laser annealing. In Figure 1b, after the laser treatment, the contact angle increased from 43.52° to 56.98°, indicating that the film became less hydrophilic. When combined with ammonia vapor (AV) atmosphere treatment, the contact angle decreased to 53.18°, indicating that AV atmosphere treatment can adjust the surface structure of the film and reorganize the surface structure after laser treatment.

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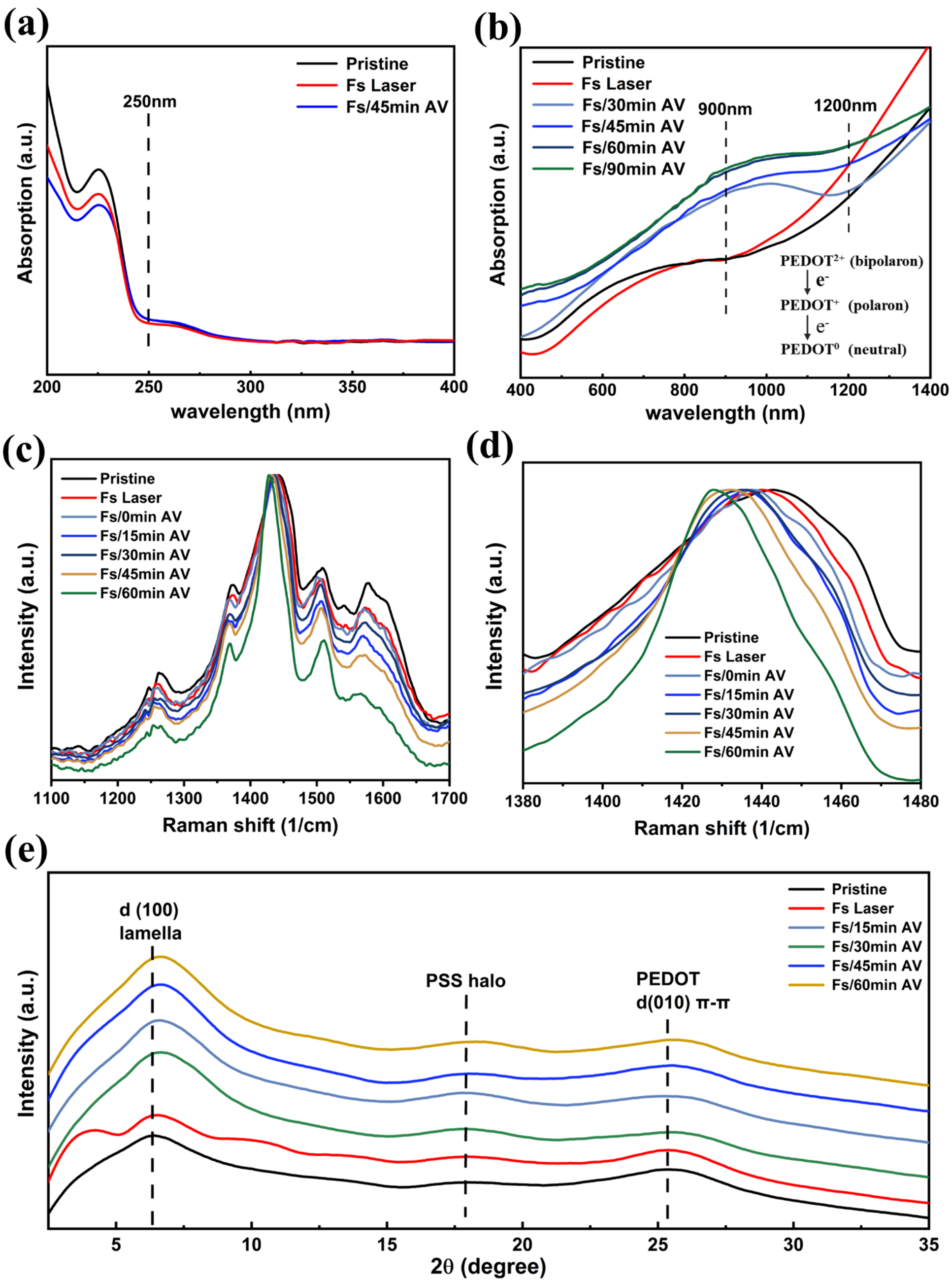

Figure 3. (a) UV absorption spectrum, (b) visible−NIR absorption spectrum, and (c) Raman spectra of pristine, laser treated, and laser and different AV atmosphere treated PEDOT:PSS films. (d) Magnified Raman spectra at 1380−1480 cm^{-1} and (e) XRD spectra of PEDOT:PSS thin films before and after treatment.

The SEM images are presented in Figure 1c for the pristine film, laser-treated film, and PEDOT:PSS film prepared by the two-step process of laser treatment followed by treatment in an AV atmosphere. Compared with the laser-treated film, after AV atmosphere treatment, the processing marks and pores on the film became inconspicuous, which confirmed that the AV atmosphere treatment can have a repairing effect on the surface of the film.

To characterize the morphological evolution more precisely, AFM measurements were performed to investigate the changes in the film morphology in detail. Figure 2 shows the morphologies and phases of the PEDOT:PSS films. The bright region corresponds to the distribution of PEDOT, while the dark region represents the region of PSS.[34] The original PEDOT:PSS film was composed of PEDOT-rich core and PSS-rich shell.[35] The morphology of the PEDOT:PSS film was characterized by clear particle boundaries, and the root-mean-square roughness (R_q) was 2.01 nm (Figure 2a). The R_q increased from 2.01 to 2.49 nm after laser treatment (Figure 2b) and decreased to 2.19 nm after further AV atmosphere treatment (Figure 2c). The PEDOT:PSS films treated with laser and AV atmosphere showed PEDOT-rich regions through an increase in the bright regions. The results show that although laser treatment can remove the PSS chains, it damages the surface of the film. However, after the co-treatment with AV atmosphere, the PSS-enriched regions are further removed, and it has a repairing effect on the surface of the film, promoting the rearrangement of the film's conductive

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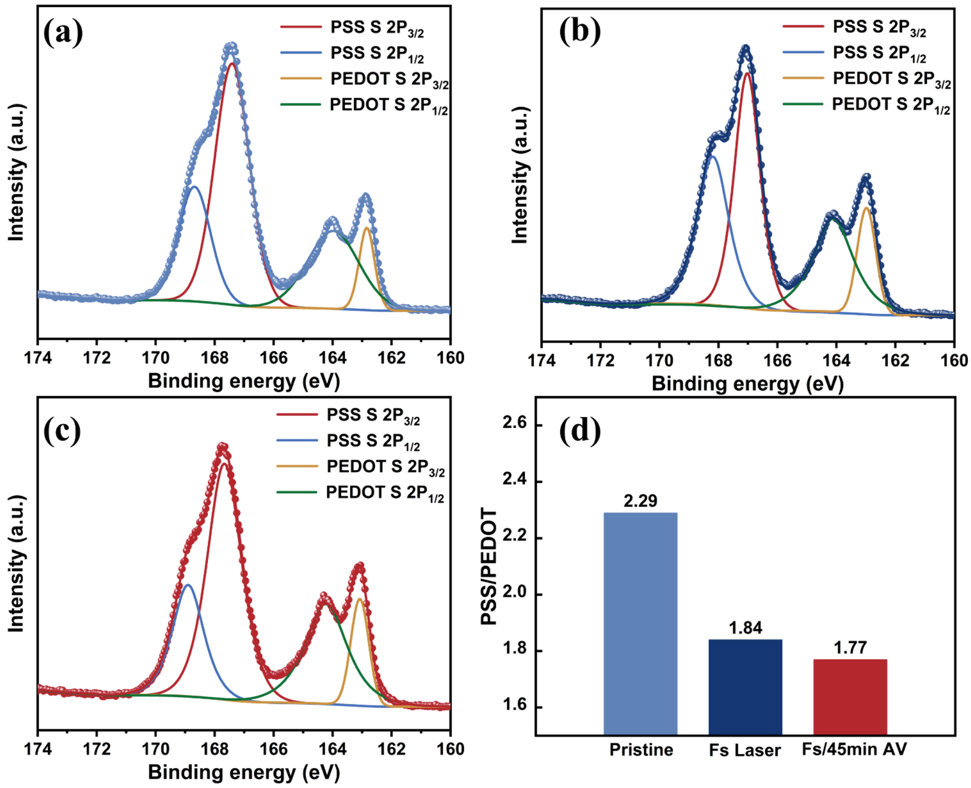

Figure 4. XPS spectra of (a) pristine PEDOT:PSS, (b) FS Laser PEDOT:PSS, and (c) FS/45 min AV PEDOT:PSS. (d) Ratio of PSS/PEDOT before and after the treatment of the PEDOT:PSS films.

structure and forming a film morphology with more uniform bright areas. The surface morphology of the original film did not show any obvious particles (Figure 2d), indicating that the PEDOT chains were fully mixed with the PSS chains and that the film was largely covered by the PSS-rich region. Evident grains appeared in the surface morphology after laser and laser-ammonia synergic treatment (Figure 2e), which were attributed to the interconnection of PEDOT-rich grains due to the removal of PSS chains, resulting in a strong phase separation between the PSS shell and PEDOT core. However, there are obvious processing marks on the surface of the film that damage the film surface. After AV atmosphere collaborative treatment, the surface of the film was repaired, and thus the phase diagram becomes more uniform (Figure 2f). These results indicate that laser irradiation and AV atmosphere treatment promote the elimination of PSS chains and the aggregation of PEDOT, and AV atmosphere treatment can repair the surface of film weaving.

To investigate the effect of the AV atmosphere treatment on the doping level of the PEDOT:PSS films, the absorption spectra of the PEDOT:PSS films before and after treatment were characterized. For clarity, the film without treatment was labeled "Pristine," the sample treated only with femtosecond laser was labeled "FS Laser", and the samples labeled with femtosecond laser and ammonia vapor atmosphere were labeled "FS/AV". "FS/15min AV," "FS/30min AV," "FS/45min AV," and "FS/60min AV" represent the PEDOT:PSS films treated in 0.3 M ammonia for 15, 30, 45, and 60 min, respectively. As shown in Figure 3a, FS/45min AV films show decreased absorption strength below 250 nm compared to the "FS Laser" films, indicating a decrease in PSS content in the films.[35,36] The neutrals, polaron, and bipolar polaron of PEDOT chain correspond to different absorption wavelengths in UV−vis−NIR spectra. As shown in Figure 3b, compared with the FS laser films, the absorption intensity of the FS/AV films was significantly enhanced at 900 nm and decreased at 1200 nm, showing that the PEDOT chain is dedoped. This means that AV atmosphere treatment results in a decrease in PEDOT chain oxidation.[37] At the same time, it was found that the absorption intensity change became more pronounced as the time of ammonia treatment increased, which indicates that the AV atmosphere treatment could help regulate the doping level of PEDOT.

Furthermore, the vibrational patterns of PEDOT chains can be revealed by Raman spectroscopy (Figure 3c,d). The peak of the original PEDOT:PSS film appears at 1442 cm^{-1}. After laser annealing treatment, the peak red-shifts to 1435 cm^{-1}, corresponding to the transformation of the PEDOT molecule from aromatic structure to quinoid structure.[20,38] With the increase of the ammonia treatment time, the red shift of characteristic peaks is more obvious, and the peak becomes sharper. After treatment with ammonia for 60 min, the red shift of the characteristic peak to 1428 cm^{-1} indicates that the molecular configuration of PEDOT changed further, which promoted the extension of conjugated chain. Combined with absorption spectroscopy, it is shown that the dopant level of PEDOT decreases and the polaron and neutral state of the polymer chain changes after AV atmosphere treatment, increasing the polaron concentration.

To study the variations in the crystallinity and $\pi-\pi$ stacking distance of the PEDOT films and their postprocessing effects, the samples were tested using XRD (Figure 3e). For the original PEDOT:PSS, there were three peaks near $2\theta = 6.42°$, 17.89°, and 25.29°. The peak at $2\theta = 6.42°$ is attributed to PEDOT on the PSS chain. The peaks at $2\theta = 17.89°$ are related to the amorphous PSS halo, whereas the (010) peak at

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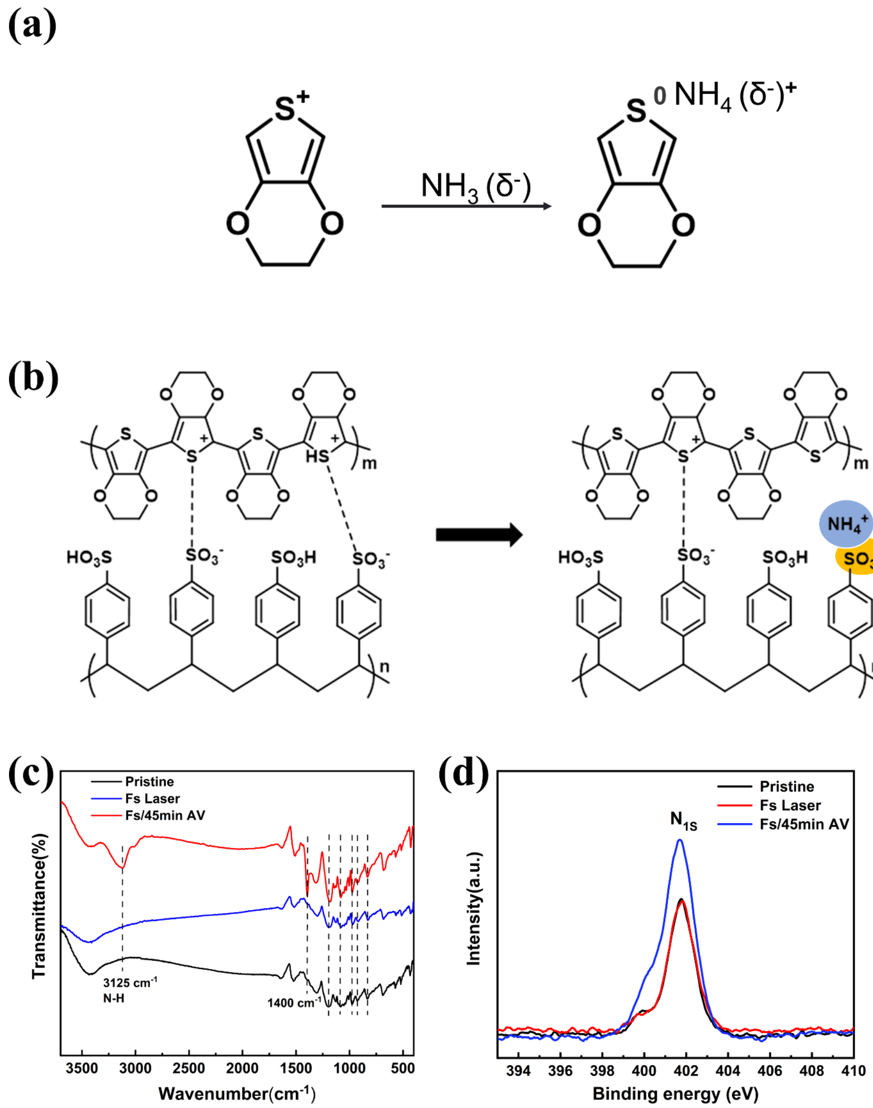

Figure 5. (a) Schematic diagram of hole and electron binding between PEDOT and NH_3. (b) Schematic diagram of PSS^- and NH_4^+ interaction process. (c) FTIR absorption spectra and (d) XPS absorption spectra of pristine films, laser treatment, and laser and ammonia atmosphere treatment PEDOT:PSS films.

2θ = 25.29° is related to the $\pi-\pi$ stacking of the PEDOT domain.[12,39] The (010) peak shifts to a higher angle with the increasing of the ammonia treatment time, attributed to the removal of PSS after laser annealing. This removal weakens the interaction between PEDOT and PSS, which brings the aggregation of PEDOT chains and the further decrease of the $\pi-\pi$ chain distance within the PEDOT domain. The peaks of (010) moved mostly toward the high angle after 45 min of ammonia treatment, which was probably due to NH_4^+ entering and PSS^- interaction to form ammonium salt. The interaction force between PEDOT and PSS decreased, resulting in the rearrangement of polymer chains and thus achieving a state of tight alignment.

X-ray photoelectron spectroscopy (XPS) was carried out to investigate the chemical composition changes in polymer PEDOT:PSS films. The binding energies of sulfur atoms in PEDOT are below 166 eV, while those in PSS exceed this threshold.[40] The S (2P) spectra of "Pristine", "FS Laser", and "FS/45min AV" samples and their deconvolution are shown in Figure 4a−c, where each S (2p) peak can be resolved into two components, $2P_{3/2}$ and $2P_{1/2}$. Two XPS bands of 168.45 and 169.7 eV are the S (2p) bands for sulfur signals from PSS,[41] whereas two XPS bands of 164.15 and 165.45 eV are the S (2p) bands for sulfur signals from PEDOT.[42] As shown in Figure 4d, after laser annealing, the PSS/PEDOT ratio decreased from 2.29 to 1.84 due to the removal of PSS, whereas the PSS/PEDOT ratio decreased further after subsequent AV atmosphere treatment. The results show that the AV atmosphere can reduce the interaction force between PEDOT and PSS, and then promote a decrease of PSS content.

Next, we illustrate the mechanism of ammonia atmosphere treatment on PEDOT:PSS films. The ammonia solution becomes unstable and decomposes into ammonia gas and water when heated to 60 °C. At the same time, ammonia gas dissolved in water reacts with water to produce ammonium cations NH_4^+ and OH^-. The formula is as follows:

$$NH_3{\cdot}H_2O \rightleftharpoons NH_3\uparrow + H_2 \quad (1)$$

$$NH_3{\cdot}H_2O \rightleftharpoons NH_4^+ + OH^- \quad (2)$$

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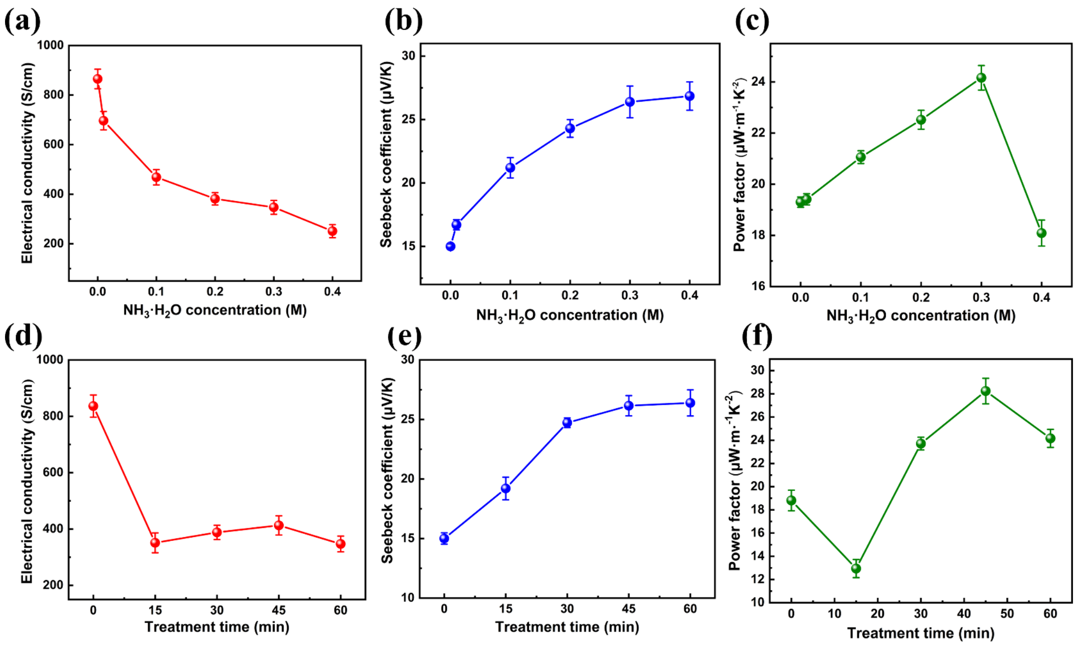

Figure 6. (a−c) TE properties of PEDOT:PSS films at ammonia concentrations: (a) σ; (b) S; (c) PF. (d−f) TE properties of laser-treated PEDOT: PSS films treated for different times in an atmosphere of 0.3 M ammonia concentration: (d) σ; (e) S; (f) PF.

Therefore, ammonia solution is a mixture containing molecules (NH_3, $NH_3{\cdot}H_2O$, and H_2O) and ions (NH_4^+ and OH^- and trace amounts of H^+). The influence of AV atmosphere on the structure and doping of PEDOT:PSS films is analyzed according to the formula and the above-mentioned characterization. In the ambient environment of NH_3, electrons undergo a transfer process from the NH_3 molecules to the surface of PEDOT, subsequently binding to the available holes within the PEDOT matrix, resulting in a change in the oxidation state of the PEDOT, i.e, a dedoping reaction (Figure 5a).[43] Thus, the binding of electrons and holes in the PEDOT leads to the formation of a neutral polymer backbone (formula 3), and the number of charge carriers decreases. This phenomenon is widely used to explain the effect of acidic/basic solvents on the thermoelectric properties of polymers (doping/dedoping process).[35]

$$\mathrm{PEDOT}^+ + NH_3 \rightarrow \mathrm{PEDOT}^0 + NH_4^+ \tag{3}$$

In addition, it is inferred from the possible protonation in PEDOT:PSS solution that NH_3 is absorbed during the reaction with PEDOT^+,[43] leading to the formation of NH_4^+ by combination with a proton. This process allows ammonium ions to enter the film as a new counteragent to PSS^-, forming the NH_4^+:PSS^- complex, which could impair the interaction between PEDOT and PSS (Figure 5b). Therefore, the molecular conformation of PEDOT changes from coiling to expanding coil or linear configuration. The carrier concentration of PEDOT was decreased by the interaction between PSS^- with NH_4^+.

To further demonstrate the above theoretical mechanism, we used Fourier transform infrared (FTIR) spectroscopy to investigate the composition of the PEDOT:PSS films after laser and AV atmosphere treatments (Figure 5c). We found that polymer films treated in an AV atmosphere show additional strong absorption near 3150 cm^{-1} and 1400 cm^{-1}, corresponding to N−H stretching and deformation vibrations of ammonium group $\delta(NH_4^+)$,[44] which are invisible in the original PEDOT:PSS. It is proved that ammonium salt PSS-NH_4^+ was formed when PEDOT:PSS is treated in AV atmosphere. The composition of the PEDOT:PSS films was characterized by XPS, where an obvious N 1S peak was observed (Figure 5d), but this peak was almost negligible in the original PEDOT:PSS films. Through the interaction of PSS^- with NH_4^+, the interaction force between PEDOT and PSS was reduced; thus, the flatness of PEDOT chains and the crystallinity of the films were improved.

We conducted tests on the thermoelectric properties of these thin films, as shown in Figure 6a−c, and the S of the films treated in an AV atmosphere was significantly higher than that of the films treated only by laser annealing. When the concentration of ammonia increased from 0.01 M to 0.4 M, AV atmosphere treatment time is maintained for 60 min, the S increased from 15 $\mu V\ K^{-1}$ to 26.85 $\mu V\ K^{-1}$, but the σ decreased rapidly, from 836.23 S cm^{-1} to 251 S cm^{-1}. The decrease of σ may be related to the change of polymer oxidation level induced by ammonia treatment.[35,45] At the optimal ammonia concentration of 0.3 M, the PF is improved up to 24.16 $\mu W{\cdot}m^{-1}{\cdot}K^{-2}$, indicating the promotion of ammonia atmosphere treatment in S after the laser annealing processes.

It has been reported that TE performance gains in PEDOT:PSS are dependent on postprocessing time.[35] To realize the effect of the post-treatment time on the thermoelectric properties, we treated the film after laser annealing for 15, 30, 45, and 60 min in the 0.3 M AV atmosphere. As shown in Figure 6d−f, S is insensitive to long-term ammonia post-treatment at around 26.39 $\mu V\ K^{-1}$. In contrast, σ is very sensitive to the AV atmosphere treatment time. Under long-term AV atmospheric treatment, σ decreased significantly to 347 S cm^{-1} after 60 min of ammonia treatment. The electrical conductivity first decreased rapidly with the AV atmosphere treatment time, followed by a gradual decrease after a certain period. At a concentration of 0.3 M ammonia for 45 min, the S increases to 26.15 $\mu V\ K^{-1}$, the σ decreases to 413.19 S cm^{-1},

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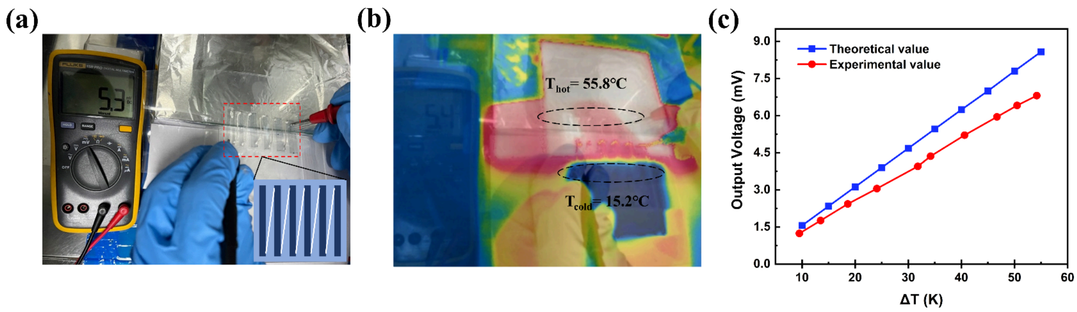

Figure 7. (a) Output voltage of a thermoelectric leg in series in the presence of a temperature difference (inset: series schematic). (b) Infrared thermography of the TE device under test. (c) Temperature difference between the two ends of a thermoelectric device versus output voltage.

and the optimal PF obtains 28.24 μW·m^{-1}·K^{-2}. The positive effects result from the partial removal of surface-insulating PSS from PEDOT:PSS films through laser treatment and the aggregation of PEDOT chains exposed to the surface after ammonia treatment, both of which enhance carrier transport capability. Simultaneously, the laser treatment etched parallel traces on the surface of the film, diminishing its smoothness and prompting the rearrangement of molecular conformations at the film surface. Consequently, the contact angles of the pristine, femtosecond laser-treated, and femtosecond laser combined with AV atmosphere-treated films exhibited different behaviors (Figure 1b).

To prove that the PEDOT:PSS/AV films are suitable for thermoelectric applications, thermoelectric devices were fabricated based on the modified films for thermoelectric legs.[46,47] As shown in Figure 7a, the device consists of six thermoelectric legs (PEDOT:PSS/AV film) fixed to a polyethylene (PE) film substrate with an inset schematic illustration of six thermoelectric legs in series. The thermoelectric leg size was 30 mm × 5 mm, and the thermoelectric leg was connected using a conductive silver slurry. One side of the prepared thermoelectric device is placed at one end of the heating platform. After the temperature of both sides of the thermoelectric legs is stabilized, the temperature of the hot and cold ends is 55.8 and 15.2 °C, respectively. The temperature difference between the two ends is 40.6 °C (Figure 7b). Then, the meter measures the output voltage of the TE leg in series at 5.3 mV, based on the formula[48]

$$V_0 = \Delta T \cdot S \cdot N \tag{4}$$

where V_0 represents the output voltage of the thermoelectric device, ΔT represents the temperature difference between hot and cold ends, and N represents the number of thermoelectric legs. The theoretical output voltage of 6.3 mV is slightly higher than the experimental output voltage, which may be due to the contact resistance in the equipment and the heat loss to the environment.

We then placed one side of the homemade TE device on a hot plate and the other side in the environment. The temperature difference between the two ends of the TE device was controlled by adjusting the temperature of the hot plate, and the output voltage of the TE devices under various temperature differences was measured.[49] Figure 7c shows the relationship between the output voltage (V_0) of the TE device and the temperature difference (ΔT) in the temperature range of 0−60 K. The red and blue curves represent the measured output voltage and the theoretical value, respectively. The theoretical values were higher than the measured values. This substrate in the TE film acts as a barrier to the heat source, resulting in the measured temperature being lower than the actual temperature of the heat source. The experimental results confirm that the two-step treatment of films exhibits practical potential for application in flexible thermoelectric devices.

4. CONCLUSION

We proposed a two-step approach combining laser annealing and AV atmosphere treatment processes to fabricate high-quality thermoelectric films with contactless, green, and nondestructive preparation conditions that do not affect the integrity of the film and do not require the addition of chemical dopants. First, PEDOT:PSS films were treated with a femtosecond laser to reduce the insulating PSS content, thereby transforming the PEDOT chain conformation from a class of benzoid type to a class of quinoid type, to improve its conductivity, but the S coefficient decreased. Subsequently, through the treatment in an AV atmosphere, ammonia supplies electrons to the surface of PEDOT, facilitating the occurrence of the dedoping reaction. The combination of NH_4^+ and PSS^- weakened the interaction force between PEDOT and PSS and increased S, thereby enhancing its overall thermoelectric performance. Under optimal conditions, the S of the film is 26.15 μV K^{-1}, and σ and PF are improved up to 413.19 S cm^{-1} and 28.24 μW·m^{-1}·K^{-2}, respectively. Furthermore, a flexible TE device was successfully fabricated using the treated film and generated a voltage of 5.3 mV at the temperature difference of 40.6 °C. This two-step strategy holds great potential for the processing of personal, portable, and flexible thermal modules.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsapm.5c01737.

Ultraviolet absorption spectra, Raman spectra, and XRD spectra of both the femtosecond laser-processed and the original PEDOT:PSS films; thermogravimetric analysis and absorption spectra of the original PEDOT:PSS film; variations in thermoelectric performance before and after femtosecond laser processing (PDF)

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Author Contributions

‡Shihao Zhang and Yunfan Yue contributed equally. Shihao Zhang: data curation, methodology, formal analysis, investigation, writing—original draft. Yunfan Yue: data curation, investigation, writing—review and editing. Jiqiang Qiao: conceptualization, resources. Zhuoran Chen: formal analysis. Xiangyu Chen: writing—review and editing, resources. Zhongle Zeng: writing—review and editing, resources. Jiakang Zhou: writing—review and editing, resources. Zihan Zhang: writing—review and editing, resources. Huan Wang: formal analysis, validation, writing—review and editing. Niannian Yu: formal analysis, validation, writing—review and editing. Xuewen Wang: funding acquisition, project administration, writing—review and editing, supervision. Qingjie Zhang: formal analysis, supervision.

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Notes

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